Prompt & RAG Experimentation Framework

Contoso Health System (CHS) — Clinical Knowledge Assistant

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1. Purpose, Scope and Environment Tags

This framework governs how prompt and RAG changes to the CHS Clinical Knowledge Assistant are proposed, tested, approved, promoted and rolled back. Every production answer traces to a **release tuple** (prompt, index, embedding and model versions), and every promotion is reconstructable from the audit trail. Environment values appear as `{{TAGS}}`, resolved per environment from Key Vault `{{KEYVAULT_URI}}` and a pinned lockfile.

Tag	Resolves to	Used for
`{{FOUNDRY_PROJECT_ENDPOINT}}`	https://<resource>.services.ai.azure.com/api/projects/<project>	Evaluation runs; judge model `{{JUDGE_DEPLOYMENT}}`
`{{AOAI_ENDPOINT}}`	Foundry model inference	Generation and embeddings
`{{SEARCH_ENDPOINT}}`	https://<service>.search.windows.net	Per-domain indices, release aliases
`{{LANGUAGE_ENDPOINT}}`	Azure AI Language PII detection (PHI domain)	PHI scans
`{{AUDIT_STORAGE_ENDPOINT}}`	Immutable (WORM) blob container	Audit records
`{{APPINSIGHTS_CONNECTION}}`	Application Insights	Telemetry, drift alerts
`{{SDK_AI_EVALUATION}}`, `{{SDK_AI_PROJECTS}}`, `{{SDK_SEARCH}}`, `{{SDK_OPENAI}}`, `{{SDK_IDENTITY}}`, `{{SDK_TEXTANALYTICS}}`	Pinned versions of azure-ai-evaluation, azure-ai-projects, azure-search-documents, openai, azure-identity, azure-ai-textanalytics	Reproducible runs; upgrades are change-controlled

Table 1. Environment tags (Entra ID principal tags: Table 9).

2. Risk Classification Rules

Tiers use the scenario brief's names verbatim and drive every gate, threshold and SLA:

- **Critical:** output can directly change a medication decision.
- **Medium-High:** output guides clinical action a clinician reviews first, or is patient-facing.
- **Low:** no direct clinical consequence.
- **Highest tier wins:** anything spanning tiers takes the highest (a leaflet mentioning an interaction is Critical); shared components inherit the highest tier they feed.
- **Most restrictive state:** content must satisfy the strictest of the five state pharmacy boards.

Use case	Code	Tier	Gate	Primary reviewer
Drug-interaction checks	DDI	Critical	GATE-CRITICAL-v1	Pharmacy SME + Clinical Safety Officer
Protocol lookup	PRO	Medium-High	GATE-MEDHIGH-v1	Specialty-matched Clinical SME
Patient education generation	PED	Medium-High	GATE-MEDHIGH-v1 + readability	Clinical SME + Health Literacy reviewer
Administrative guidance	ADM	Low	GATE-LOW-v1	Departmental owner (spot-check)

Table 2. Use-case risk classification.

3. PromptOps Governance Policy

3.1 Versioning and naming

- **Prompts:** `PMT-{DOMAIN}-{NNN}` with SemVer, e.g. `PMT-DDI-004 v2.1.0`. MAJOR = clinical instruction, safety rule or output format; MINOR = wording or examples; PATCH = formatting. A MAJOR change resets SME approval.

- **Embeddings and indices:** `EMB-{model}-{ver}` and `IDX-{DOMAIN}-{EMB}-{CHUNK}-{YYYYMMDD}`, e.g. `IDX-DDI-modelB-v2-SEM15-20261101`. Production reaches an index only through its release alias (`ddi-prod`).
- **Experiments:** `EXP-{YYYY}-{NNNN}`, one manifest each. Prompts and manifests live in Git, not application code.

3.2 Change control and approval gates

Every change needs a manifest, a change-control ticket and two-person pull-request review. SME and compliance approvals are required-reviewer approvals on the `{{ENV_PROD_APPROVAL}}` environment, and gates fail closed: an expired SLA escalates, never auto-approves.

Tier	Approvals before any production exposure	Exposure path
Critical	Automated gate + Pharmacy SME (Clinical Safety Officer on escalation) + Compliance	Shadow only after pharmacist sign-off, then staged
Medium-High	Automated gate + specialty SME (+ Health Literacy for PED) + Compliance	Shadow during review, then staged
Low	Automated gate alone; departmental owner spot-checks weekly	Staged

Table 3. Approval gates. Staged rollout: one site, one region, then all 12 hospitals, each after a clean monitoring window.

Critical changes enter shadow mode only after pharmacist sign-off, because the brief counts shadow mode as production exposure; the supplied workflow placed it before review. Emergency formulary recalls get an expedited pharmacy re-index but keep the Critical gate.

3.3 Authoring standards and PHI minimization

- No PHI in prompts, examples, logs or evaluation datasets; datasets are synthetic or de-identified (HIPAA Safe Harbor), PHI-scanned via `{{LANGUAGE_ENDPOINT}}` before snapshotting, and exclude 42 CFR Part 2 content.
- Clinical prompts are cite-or-abstain: answer only from retrieved passages with the passage ID, otherwise state “not found in approved sources” and refer onward.
- Responses carry a decision-support disclaimer; PED output targets a grade 6–8 reading level.
- Query logs hash user IDs and redact free text; retention follows compliance policy.

3.4 Required healthcare metadata

Group	Fields (manifest and audit record)
Identity	experiment_id, use_case, clinical_risk_tier, prompt_id, semantic_version, author, change_control_ticket
Clinical provenance	source_corpus_version, **formulary_effective_date**, **states_validated**, **regulatory_refs** (HIPAA, 42 CFR Part 2, state boards, FDA labeling)
Privacy	**phi_attestation**, **phi_scan_result**, dataset_snapshot_id
Approval and audit	gate_thresholds_ref, automated_gate_result, sme_approver_id, sme_decision + timestamp, **compliance_approver_id**, audit_token, retention_policy
Rollback	**rollback_target** (last known-good release tuple), rollback_flag, rollback_reason

Table 4. Required metadata. Bold fields extend the supplied manifest skeleton.

3.5 Rollback triggers

Tier	Incident triggers	Owner	SLA	Review
Critical	Groundedness below 4.6 for 2 consecutive monitoring windows; any clinician-reported interaction error	GenAIOps on-call; Clinical Safety Officer notified	30 min	5 days
Medium-High	Groundedness below 4.25 for 3 consecutive windows; clinician-reported inaccurate protocol guidance	GenAIOps on-call	120 min	5 days
Low	3+ user-reported factual errors per week	GenAIOps on-call	480 min	10 days

Table 5. Rollback triggers and SLAs.

Rollback repoints the affected domain’s release alias to `rollback_target`; domain-segmented indices mean a pharmacy rollback leaves other domains untouched. On-call engineers activate the deploy permission just-in-time through Privileged Identity Management, and each rollback is audited.

4. RAG Experiment Matrix

Each experiment changes one dimension against the production baseline, offline, on a frozen PHI-free dataset. Priority weighs expected impact against the safety cost of a regression; P1 runs first.

ID	Dimension	Variants (baseline first)	Scope	Impact	Safety risk	Priority
R1	Chunking: drug monographs	Fixed 512 tokens/10% overlap vs. interaction-atomic (one drug pair per chunk)	DDI	High	Critical	P1
R2	Chunking: protocols	Fixed 512/10% vs. semantic, heading-aware, 15% overlap	PRO	High	Medium-High	P1
R3	Retrieval mode	Dense vs. hybrid dense + BM25 (exact drug names, doses)	DDI, PRO	High	Critical	P1
R4	Prompt grounding	Cite-or-abstain v1 vs. v2 (stricter refusal)	DDI, PRO	High	Critical	P1
R5	Embedding model	modelA-v1 vs. modelB-v2 (full re-index per domain)	All	High	Critical	P2
R6	Retrieval depth	top_k 3 / 5 / 8 with reranker-threshold sweep	All	Medium	Medium-High	P2
R7	Context window	4k vs. 8k tokens of retrieved context	PRO, PED	Medium	Medium-High	P2
R8	Readability prompt	Grade 8 vs. grade 6 instruction	PED	Medium	Medium-High	P2
R9	Index strategy	Per-domain vs. unified (offline only: unified loses independent freshness and rollback)	All	Low	Critical	P3

Table 6. RAG experiment matrix.

4.1 Reproducibility checklist

- Manifest committed and linked to its ticket before the run.
- Immutable, PHI-scanned `dataset_snapshot_id`; fixed `random_seed`; temperature 0 for evaluation runs.
- Pinned judge model (`evaluator_model_version`); evaluators run three times and are averaged to absorb judge variance.
- SDK versions from lockfile tags; chunker commit SHA, index, embedding and formulary versions recorded.
- Rerunning from the manifest alone reproduces gate scores within ±0.1; raw outputs stored in `{{AUDIT_STORAGE_ENDPOINT}}`.

5. Evaluation Framework

5.1 Metrics and automated gates

The runner calls azure-ai-evaluation (`{{SDK_AI_EVALUATION}}`) against `{{FOUNDRY_PROJECT_ENDPOINT}}`, scoring on the native 1–5 scale with gates from evaluation_gates.yaml.

Use case	Gate	Quality thresholds (1–5)	Safety and domain checks	On fail
DDI	GATE-CRITICAL-v1	Groundedness ≥4.6 (mean and every response); relevance ≥4.3; retrieval ≥4.2	Content safety: any flag; drug_interaction_accuracy: 0 missed, 0 fabricated; PHI leak: 0	Block
PRO	GATE-MEDHIGH-v1	Groundedness ≥4.25; relevance ≥4.0; coherence ≥3.8; completeness ≥3.8 (preview, provisional)	Content safety; citation present; PHI leak: 0	Hold for SME
PED	GATE-MEDHIGH-v1	As PRO	As PRO, plus readability grade ≤8 (custom)	Hold for SME
ADM	GATE-LOW-v1	Relevance ≥3.6; fluency ≥3.6	Content safety; PHI leak: 0	Block

Table 7. Metrics by use case. Content safety and PHI checks run at every tier (extends the sample YAML).

Built-in groundedness checks support by retrieved context, not pharmacological correctness: a stale monograph can be perfectly "grounded". Hence the SME-curated drug_interaction_accuracy evaluator blocks independently, and R1 and R3 also require labelled recall@k of 1.0 on known interaction pairs.

5.2 Human-in-the-loop triggers and SLAs

Tier	Triggers	Reviewer	SLA	Action
Critical	Any safety flag or custom-evaluator failure; any response below 4.6 groundedness; drug entity unseen in 90 days of evaluation sets	Pharmacy SME; Clinical Safety Officer escalation	4 h	Pause promotion; page via `{{PAGING_SYSTEM}}`
Medium-High	Any evaluator below threshold; groundedness drift above 0.3 vs. 30-day production baseline	Specialty SME or Health Literacy reviewer	24 h	Hold in shadow; queue
Low	Gate failure in 3 or more consecutive runs	Departmental owner	48 h	Weekly spot-check

Table 8. Review triggers. A missed SLA escalates one level and the change stays blocked.

5.3 Production monitoring

Continuous evaluation reruns each tier's evaluators on PHI-scrubbed production samples every monitoring window; `{{APPINSIGHTS_CONNECTION}}` alerts implement the Section 3.5 triggers. A freshness guard switches drug-interaction answers to abstain-and-refer whenever the pharmacy index lags the licensed formulary, until re-indexing passes the Critical gate. Clinician error reports feed review queues and rollback counts.

6. Experimentation Workflow and Roles

Figure 1 (next page) traces an experiment through callouts 1–10. Gate failures and SME rejections return to engineering as a new version in the same experiment lineage; production incidents trigger rollback, and every decision is written to the audit trail.

Role	Principal tag	Entra ID role assignments (scope)	Callouts
GenAIOps engineers	`{{GRP_GENAIOPS}}`	Foundry User (dev/test); Search Index Data Contributor (dev indices); Reader (prod)	1, 2, refine
Evaluation runner	`{{MI_EVAL_RUNNER}}`	Foundry User (eval project); Search Index Data Reader; Storage Blob Data Contributor (audit)	3, 10
Clinical SMEs	`{{GRP_PHARM_SME}}` `{{GRP_CLIN_SME}}`	No data-plane roles; required reviewers on `{{ENV_PROD_APPROVAL}}`	5
Compliance and privacy	`{{GRP_COMPLIANCE}}`	Reader; Log Analytics Reader; required reviewer	6
Deployment pipeline	`{{MI_DEPLOY}}`	Search Index Data Contributor (prod); only principal that repoints aliases	7
GenAIOps on-call	`{{GRP_ONCALL}}`	PIM-eligible, time-bound deploy role	9
Assistant runtime	`{{MI_APP_RUNTIME}}`	Foundry User; Search Index Data Reader (read-only)	8
Audit owner	`{{GRP_AUDIT}}`	Storage Blob Data Reader (audit); Log Analytics Reader	10

Table 9. Roles and Entra ID assignments (Foundry User was formerly named Azure AI User).

Contoso Health System — Clinical Knowledge Assistant: Prompt & RAG Experimentation Workflow

Proposal → validation → promotion, with risk-tier gates, feedback loops and rollback. Numbered callouts map to sections of Dennis_PromptRAG_Framework.

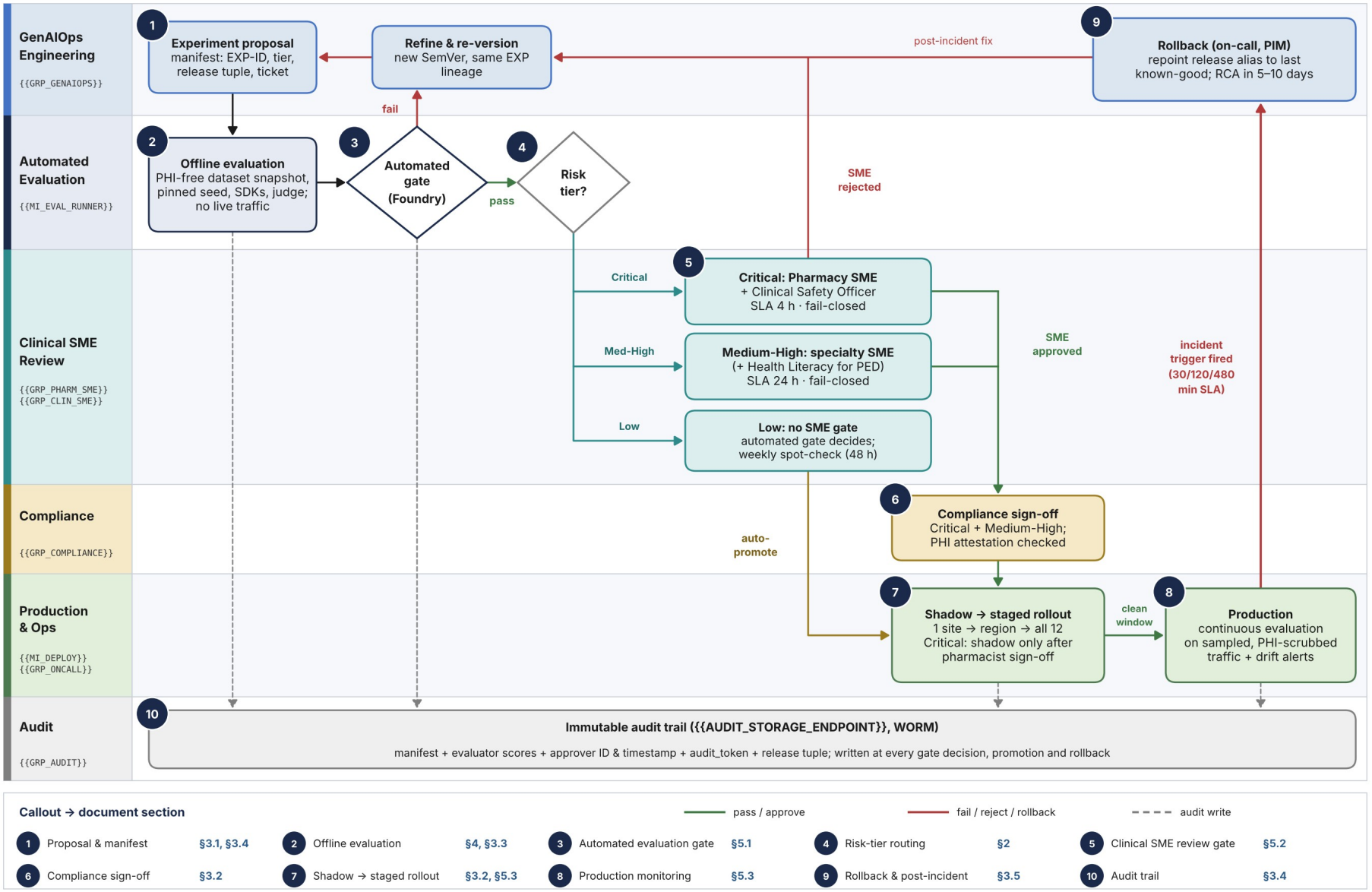


Figure 1. Experimentation workflow; numbered callouts map to document sections (also supplied as Dennis_PromptRAG_Diagram.pdf/.png).